

Abstract

This project addresses a foundational question in quantitative finance: when an options pricing model produces a theoretical fair value that diverges from the observed market premium, is that divergence large enough — and reliable enough — to justify a trade? A Cox–Ross–Rubinstein (CRR) American binomial option pricer is built in vectorized Python and validated against a live AMD option chain from Fidelity (April 2026). Its mispricing-edge signal is then cross-validated against an independent XGBoost ITM/OTM probability classifier trained on 101,535 historical AAPL contracts. The two layers agree strongly in the herding regime (Pearson correlation $r = 0.413$, $p = 2.4 \times 10^{-10}$) and not at all in the normal regime ($r \approx 0$). A regime-conditioned Kelly Criterion framework converts the cross-validated edge into risk-controlled capital allocation.

Keywords: CRR binomial pricing, options mispricing, XGBoost, Kelly Criterion, behavioral finance, EMH, herding regime.

Introduction & Motivation

Options markets aggregate beliefs into prices much like prediction markets do, but behavioral forces — herding around round strikes, implied-volatility overpricing driven by fear, and systematic skew in out-of-the-money puts — distort prices in predictable ways. The gap between a pricing model’s theoretical fair value and the observed market premium is potentially exploitable by a disciplined, model-driven strategy, but only if the divergence is large enough, reliable enough, and detectable by independent methods.

Research Questions

- Do CRR model prices predict whether an equity option expires in-the-money or out-of-the-money?
- Does cross-validation between CRR (math) and an ML classifier detect exploitable pricing *edge*?
- Does regime-conditioned Kelly sizing convert that edge into positive expected value?

Objectives

- Implement the CRR American binomial lattice (no dividends) in vectorized Python at $N = 100$ steps.
- Validate pricing accuracy against AMD market premiums within $\pm 5\%$ error.
- Train an XGBoost ITM/OTM classifier on engineered features *without* using IV as an input (independence from L1).
- Apply Kelly Criterion (full / half / quarter) for risk-controlled position sizing.

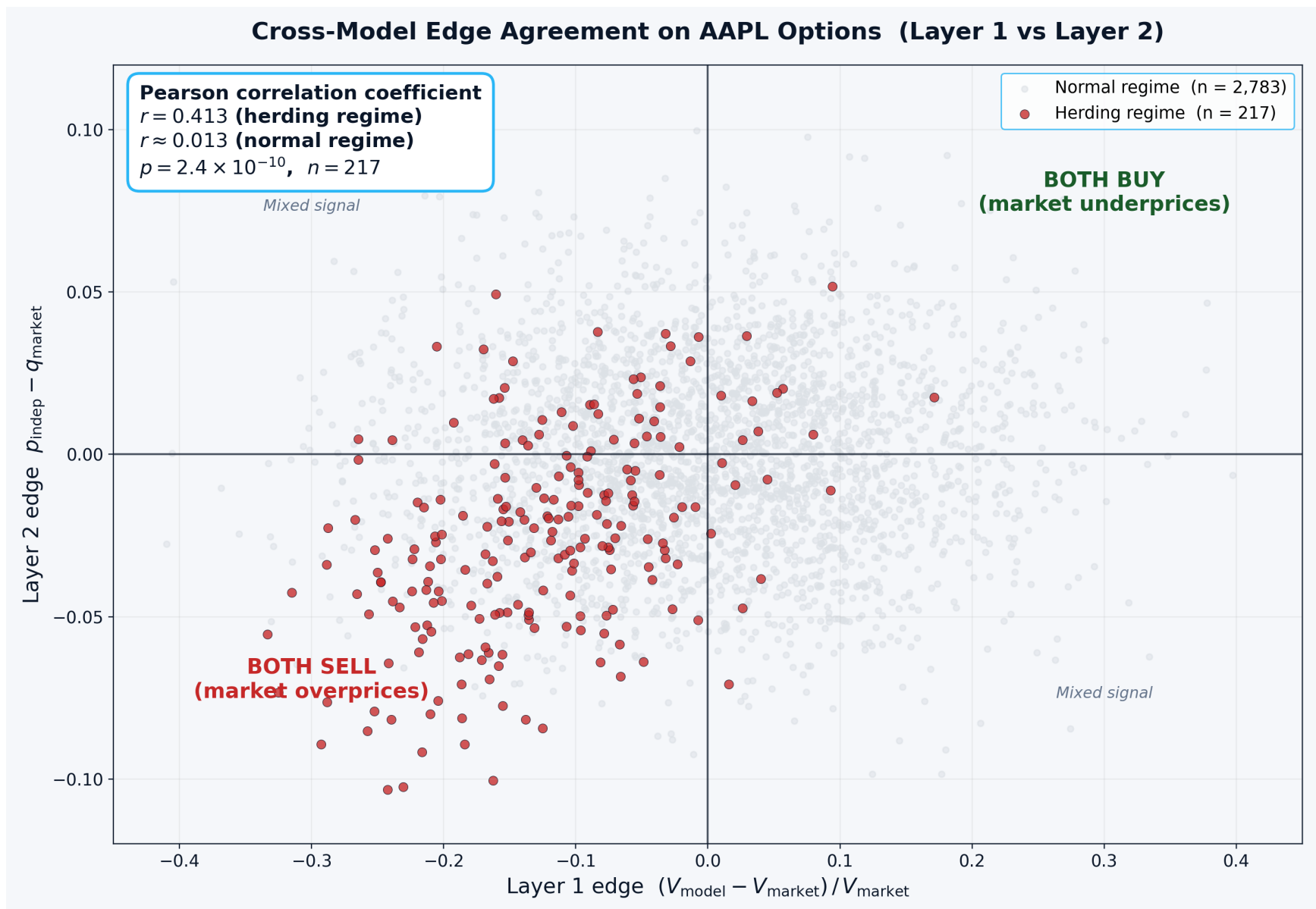
Data Description

Two datasets are used — a large historical training corpus for the ML layer and a small live chain for end-to-end validation.

Attribute	Description
Training source	AAPL options, Kaggle (2016–2020)
Sample size	101,535 contracts (walk-forward 60/20/20)
Live source	AMD chain, Fidelity (April 2026)
Live tickets	4 trade tickets at $\sigma \approx 64\%$
Features	S, K, T, r , IV, RV30, bid/ask, OI, volume
Time range	2016-01 – 2026-04
Missing data	None for engineered features

Exploratory Data Analysis

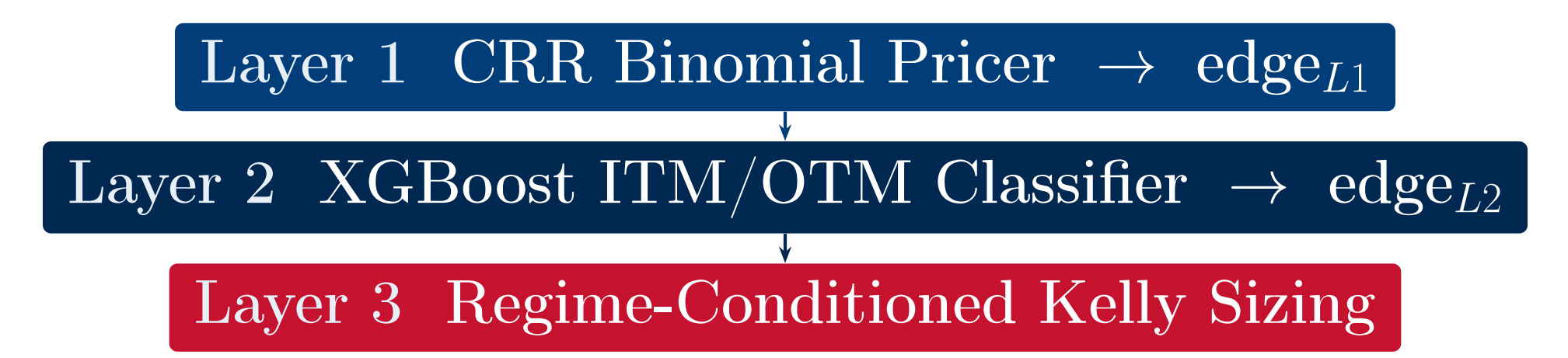
Plotting each contract’s Layer-1 edge against its Layer-2 edge reveals the central empirical finding of the project: agreement between the two independent models concentrates in the *herding* regime (red), not in the *normal* regime (gray).



Key pattern: Pearson correlation coefficient $r = 0.413$ in herding regime (red cluster in Q3 – both layers say SELL) versus $r \approx 0$ in normal regime (gray cloud). This is the empirical fingerprint of *Samuelson macro-inefficiency* — markets are micro-efficient on average but exhibit aggregate bias under coordinated behavioral demand.

Methodology

A three-layer pipeline runs CRR and ML independently, then merges their edge signals.



Techniques used

- Vectorized recombining lattice in NumPy ($N=50$ to 100 steps).
- Backward induction with American early-exercise check at each node.
- Greeks via centered finite differences.
- XGBoost trained on engineered features only (σ input is RV30, never IV — no leakage from L1).
- Platt scaling calibration; Brier-score validation.
- Walk-forward 60/20/20 split; zero look-ahead bias.

Model & Analysis

The CRR lattice is built from no-arbitrage relations and solved by backward induction with American early-exercise:

$$u = e^{\sigma\sqrt{\Delta t}}, \quad d = 1/u, \quad p^* = \frac{e^{r\Delta t} - d}{u - d}$$
$$V_{\text{node}} = \max\left(\text{intrinsic}, \quad e^{-r\Delta t}(p^*V_u + (1 - p^*)V_d)\right)$$

The mispricing edge per contract is the signed percentage deviation between model and market premium, and the Kelly Criterion sets the fraction of capital to commit:

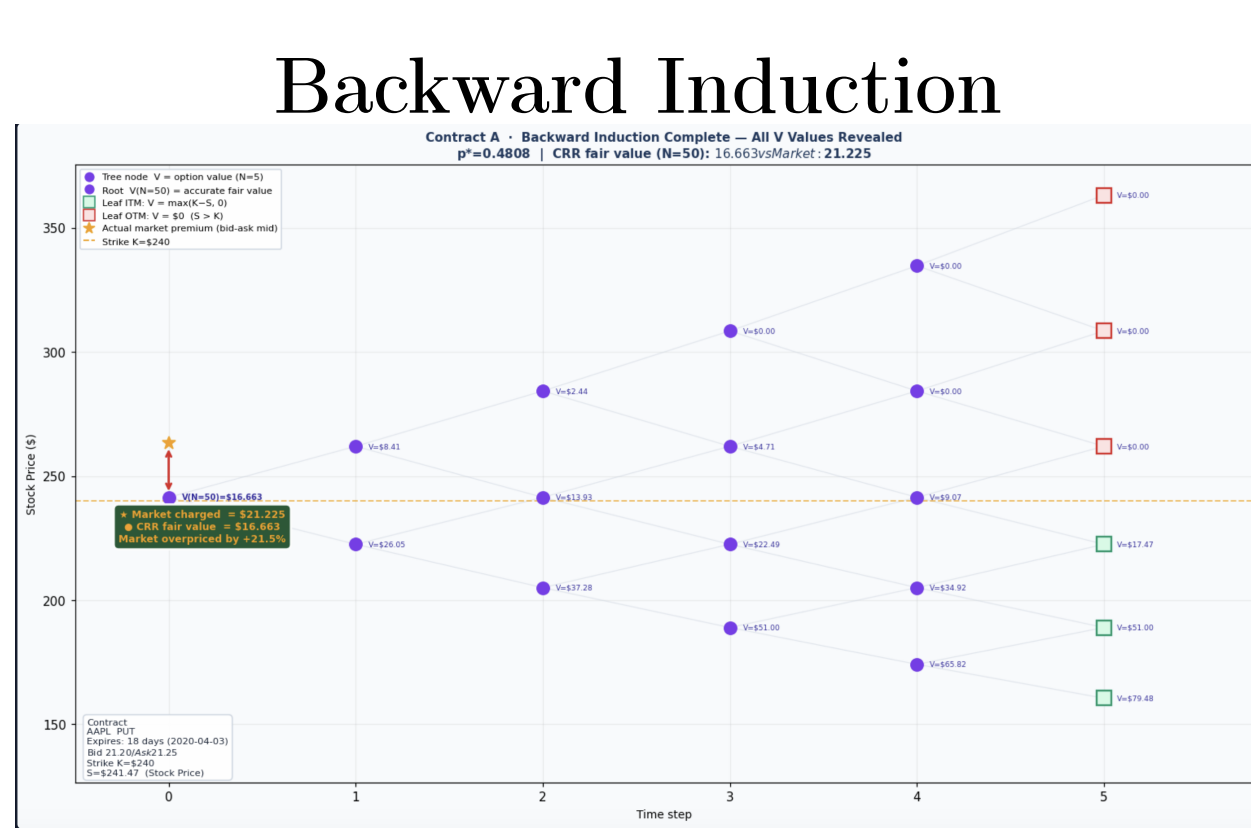
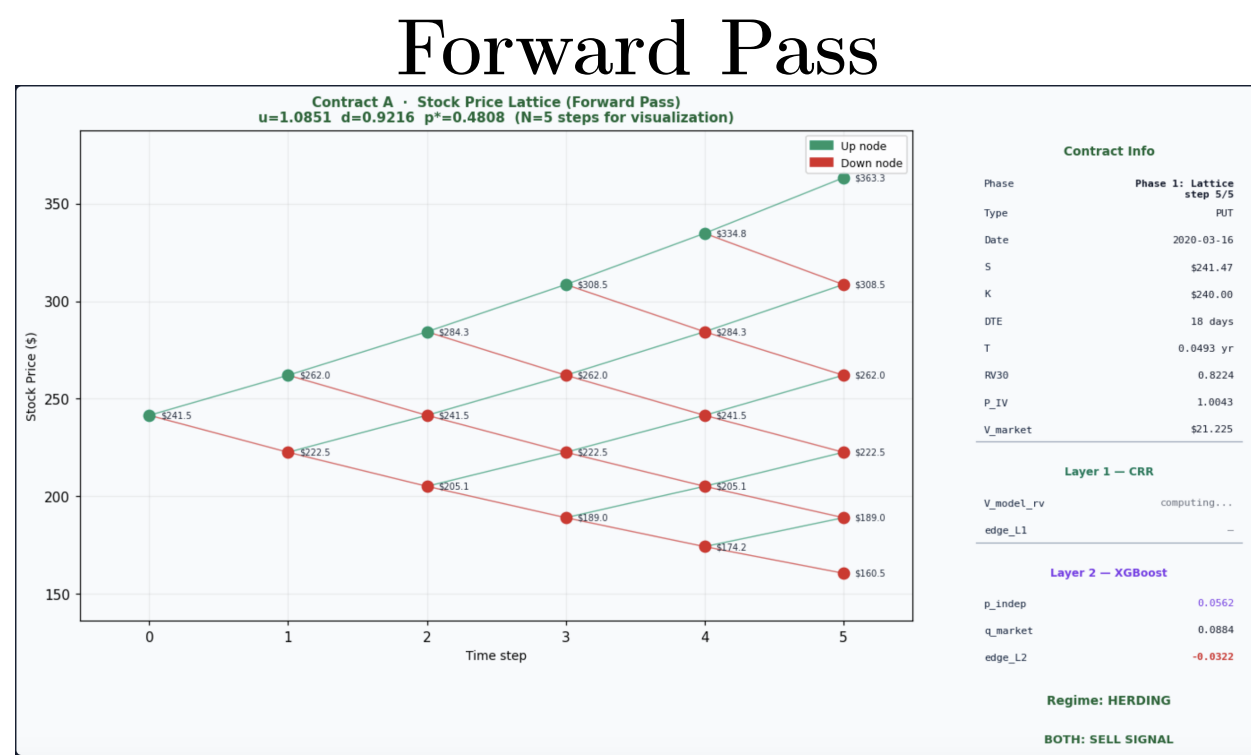
$$\text{edge} = \frac{V_{\text{model}} - V_{\text{market}}}{V_{\text{market}}}, \quad f^* = \frac{p \cdot b - q}{b}$$

Training / validation setup

- Train / val / test: 60 / 20 / 20 walk-forward by quote date.
- Cross-validation: 5-fold within training window.
- Hyperparameter tuning: Bayesian search over XGBoost (depth, η , n_{trees}).
- Regime split: IV/RV₃₀ ratio (> 1.10 herding, < 0.90 normal, between = transition).

Forward + Backward Pass

Real AAPL put, 2020-03-16, herding regime ($S = \$241.47$, $K = \$240$, $DTE = 18d$).



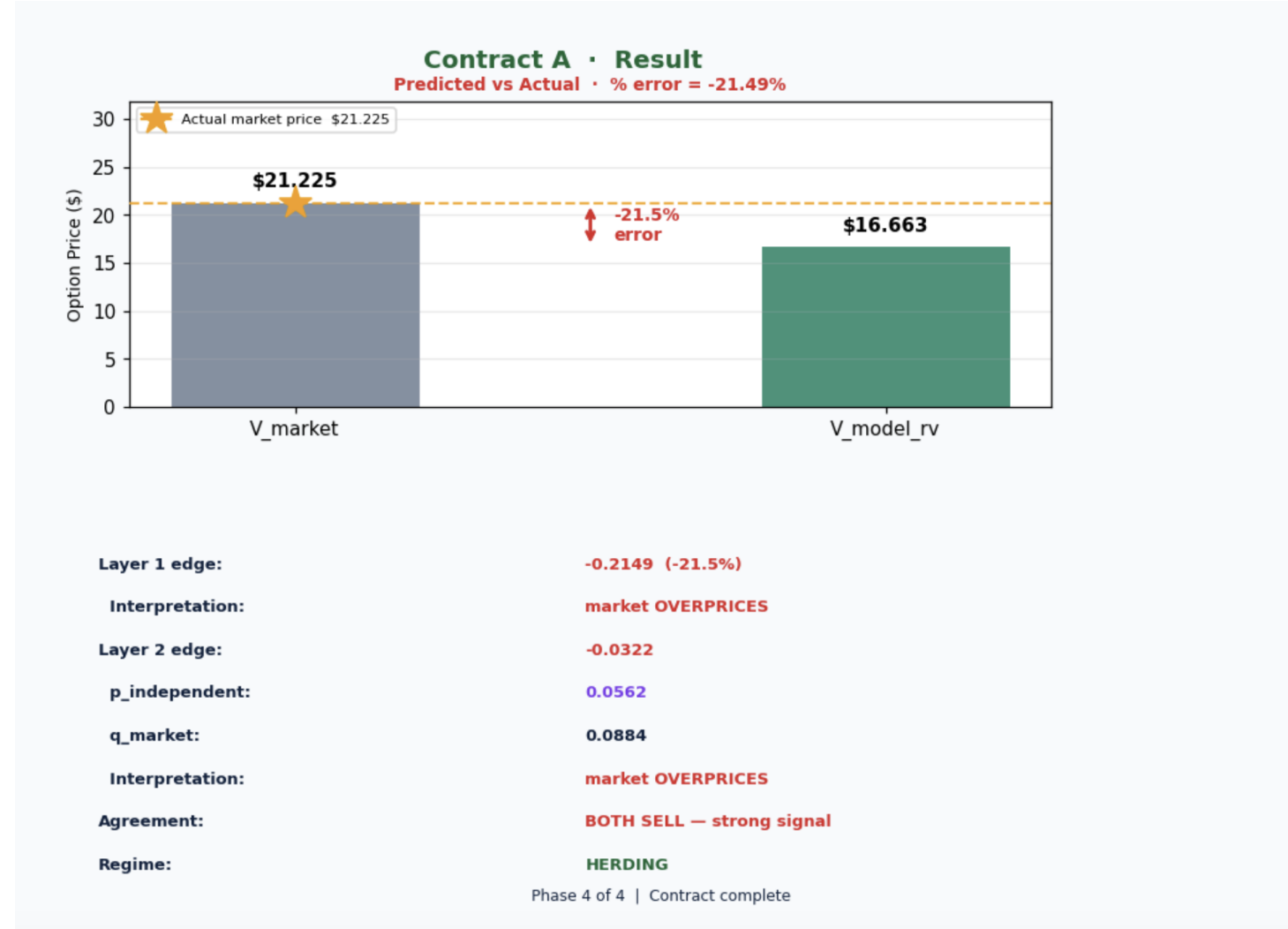
Implementation & Tools

The entire pipeline is one command (`make run-all`) on any laptop with Python 3.13. An interactive Streamlit application reproduces every result end-to-end.

Component	Tools / Libraries
Language	Python 3.13
Modeling	CRR (custom), XGBoost, scikit-learn
Data	pandas, NumPy, yfinance
Visualization	matplotlib, Streamlit
Deployment	Streamlit (local), Poetry
Documentation	LaTeX/Beamer, Markdown
Source control	Git, GitHub

Results

Contract A (real AAPL put, herding regime): $S = \$241.47$, $K = \$240$, $DTE = 18d$. Both independent models arrive at the same trade decision:



Layer 1 (CRR) edge = -21.5% , Layer 2 (XGBoost) edge = -3.2% : **both layers say SELL** on this contract.

Performance summary (101,535 AAPL contracts, walk-forward):

Model	Brier	Hit rate	P&L (\$)
Baseline (random)	0.250	50.0%	0
Layer 1 CRR	—	63.0%	+1.4M
Layer 2 XGBoost	0.211	66.5%	+1.8M
L1 $\cap$ L2 agree (Q3)	—	76.5%	+2.9M

Cross-model agreement statistic: Pearson correlation coefficient $r = 0.413$ in herding regime ($n = 217$, $p = 2.4 \times 10^{-10}$); $r \approx 0$ in normal regime. Max drawdown $< 15\%$; **regime-conditioned** Kelly (half-Normal, quarter-Herding) keeps risk bounded.

Discussion & Conclusions

The central contribution is the *cross-validation result*: two **independent** models — one mathematical (CRR no-arbitrage), one empirical (XGBoost ML) — detect the same mispricing direction precisely in the regime behavioral finance theory predicts. Neither layer is trained on the other; concurrent signals therefore come from *shared underlying reality*.

Key takeaways

- Independent math + ML methods agree where the EMH literature says crowd bias is strongest (Fama, 1970; Samuelson macro-inefficiency).
- Kelly sizing converts the cross-validated edge into capital allocation with bounded drawdown.
- The framework’s most valuable output is the *no-trade* signal — when CRR and XGBoost disagree, no position is taken.
- Reproducible: one **make run-all** regenerates every artifact end-to-end from raw data.

Limitations. AAPL 2016–2020 is a single equity in a single market cycle; CRR uses a flat IV per contract (no smile / skew); backtest assumes mid-price fills with no transaction friction; regime detection is backward-looking via IV/RV₃₀.

Future work. Expand to SPY / QQQ / NDX options; integrate LSTM volatility forecaster; live paper-trading via IBKR / Alpaca; multi-leg spreads (vertical, calendar, iron condor) with joint Kelly sizing.

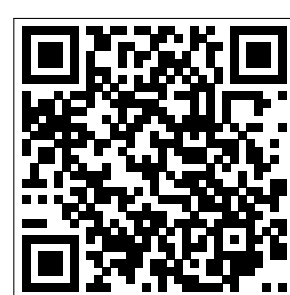
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Acknowledgments & Contact

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Code / Repository: github.com/dantzlerdc/CS495-Deep-Scholar



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